

# Changhao Li

cli911@gatech.edu · Homepage · Google Scholar · GitHub

## EDUCATION

|                                                                                                                             |                                       |
|-----------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| <b>Georgia Institute of Technology</b><br>Ph.D. in Computational Science and Engineering<br>Advisors: Bo Dai and Chao Zhang | Aug 2024 – Present<br>GPA: 4.00/4.00  |
| <b>Tsinghua University</b><br>B.Eng. in Computer Science and Technology<br>Advisors: Jianfei Chen and Jun Zhu               | Sep 2020 – Jun 2024<br>GPA: 3.89/4.00 |

## INDUSTRY EXPERIENCE

|                                                                                                                     |                     |
|---------------------------------------------------------------------------------------------------------------------|---------------------|
| <b>ByteDance Seed</b> — Student Researcher<br>Advancing the long-horizon agentic capabilities of frontier models.   | Aug 2026 – Present  |
| <b>Qwen</b> — Research Intern<br>Revisiting and simplifying on-policy distillation for LLM reasoning.               | May 2026 – Aug 2026 |
| <b>AWS AI Lab</b> — Applied Scientist Intern<br>Developed an agentic evaluation framework for deep-research agents. | May 2025 – Aug 2025 |

## SELECTED RESEARCH PROJECTS

### Agent DAgger: learning from on-policy experience

First author | Preprint, 2026

- Developed teacher-interleaved rollouts that combine on-policy state coverage with dense expert supervision, reducing covariate shift in long-horizon LLM agents.
- Achieved **27.3% / 29.8%** task resolution with 4B / 8B students on SWE-bench Verified, improving over on-policy distillation by **3.9 / 3.6 percentage points**.

### Controllable reasoning: Matryoshka Pilot and EDO

M-Pilot: co-first author, NeurIPS 2025 | EDO: first author, TMLR 2026

- Introduced **M-Pilot**, a lightweight white-box controller that steers black-box LLMs through intermediate guidance for multi-turn reasoning, planning, and personalization.
- Integrated exploration objectives into iterative DPO and GRPO with **EDO** to preserve solution diversity, improve test-time reasoning, and stabilize post-training.

### DREAM: agentic evaluation of deep-research agents

Co-first author | ACL 2026, Oral

- Developed an **agentic evaluation framework** combining tool use with query-agnostic and adaptive metrics to assess factual accuracy, temporal validity, and reasoning quality.

## OPEN-SOURCE SOFTWARE

**STACX** — Agent-training infrastructure connecting sandboxed execution, rollout scheduling, and verifier rewards, with support for multiple training techniques.

**MLE-Dojo** — Gym-style environments for iterative ML experimentation on **200+ Kaggle competitions**, with executable feedback for evaluation and training.

PUBLICATIONS & PREPRINTS

\* Equal contribution. † Equal second authorship.

[1]

DREAM: Deep Research Evaluation with Agentic Metrics

Elad Ben Avraham\*, Changhao Li\*, Ron Dorfman\*, Roy Ganz, Oren Nuriel, Amir Dudai, Aviad Aberdam, Noah Flynn, Elman Mansimov, Adi Kalyanpur, Ron Litman.

ACL 2026 – Oral [Paper]

[2]

Exploration-Driven Optimization for Test-Time Large Language Model Reasoning

Changhao Li, Yuchen Zhuang, Chenxiao Gao, Haotian Sun, Rushi Qiang, Chao Zhang, Bo Dai.

TMLR 2026 [Paper]

[3]

Matryoshka Agent: Unfolding Sub-Agents for Long-Horizon Machine Learning Engineering

Rushi Qiang, Changhao Li†, Haotian Sun†, Yuchen Zhuang, Chao Zhang, Bo Dai.

Preprint, 2026 [Paper]

[4]

Revisiting DAgger in the Era of LLM-Agents

Changhao Li, Rushi Qiang†, Jiawei Huang†, Chenxiao Gao†, Chao Zhang, Niao He, Bo Dai.

Preprint, 2026 [Paper] [Code]

[5]

Matryoshka Pilot: Learning to Drive Black-Box LLMs with LLMs

Changhao Li\*, Yuchen Zhuang\*, Rushi Qiang, Haotian Sun, Hanjun Dai, Chao Zhang, Bo Dai.

NeurIPS 2025 [Paper] [Code]

[6]

MLE-Dojo: Interactive Environments for Empowering LLM Agents in Machine Learning Engineering

Rushi Qiang\*, Yuchen Zhuang\*, Yinghao Li, Dingu Sagar V K, Rongzhi Zhang, Changhao Li, Ian Shu-Hei Wong, Sherry Yang, Percy Liang, Chao Zhang, Bo Dai.

NeurIPS 2025 [Paper] [Code]

[7]

Task-Specific Zero-shot Quantization-Aware Training for Object Detection

Changhao Li\*, Xinrui Chen\*, Ji Wang\*, Kang Zhao, Jianfei Chen.

ICCV 2025 [Paper] [Code]

[8]

Training Transformers with 4-bit Integers

Haocheng Xi, Changhao Li, Jianfei Chen, Jun Zhu.

NeurIPS 2023 [Paper] [Code]

SELECTED HONORS

|                                                                       |      |
|-----------------------------------------------------------------------|------|
| Outstanding Graduate Award, Tsinghua University                       | 2024 |
| Comprehensive Excellence Scholarship, Tsinghua University             | 2023 |
| Academic & Social Service Excellence Scholarship, Tsinghua University | 2022 |

ACADEMIC SERVICE

|                                                                        |             |
|------------------------------------------------------------------------|-------------|
| Reviewer: NeurIPS, ICLR, ICML, ACL Rolling Review, TMLR.               |             |
| Teaching Assistant: CX4240 – Computing for Data Analysis, Georgia Tech | Spring 2026 |

TECHNICAL SKILLS

Languages & frameworks: Python, C++, PyTorch, CUDA.

Systems experience: Implemented CUDA kernels for 4-bit transformer training (NeurIPS 2023).